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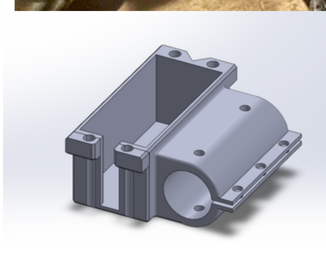
PORTFOLIO TEST VIDEOS ALBUM: https://photos.app.goo.gl/AKPdssPic5SE3iX16

EXIA LABS- ARGUS AUTOMATED MILITARY VEHICLE

The goal was to deliver a field-ready autonomous vehicle for the NATO xTech Military Competition in Germany; the plan of action was converting a civilian ATV into a 200+ kg payload carrier that could navigate unpredictable terrain without any human input, utilizing a 2D LiDAR sensor and ai to avoid objects.

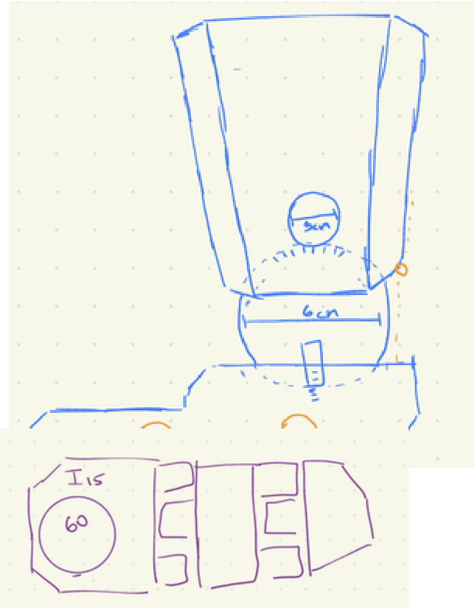
- Performed complete teardown and reverse engineering of a 2007 Suzuki KingQuad to isolate drivetrain, throttle, and braking interfaces
- Removed manual turning system to integrate autonomous gearbox to operate front-wheel turning axle
- Repaired and restored critical subsystems, including braking components, wiring harnesses, and engine air intake filters

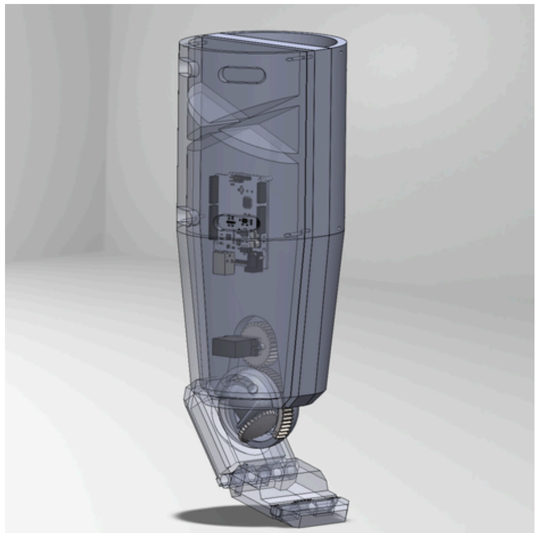
- Compared moment calculations with shear force limit of the steel shaft to determine the required supports locations (calculations required for documentation)
- Welded and angle-grinded steel frame to hold 200kg weight at any location of the designed platform
- Modelled and produced prints to mount and automate gas throttle through a micro controller and arduino board



- Utilized ODrive motors, CNC'ed steel gears and a 3D printed gear box for front wheel rotation turning
- Repaired suspension system and replaced suspension springs to work in more snowy/rocky environments
- Built a steel shell for the internal frame and electronics, bolting and welding the exterior with specific gaps for air circulation

CURRENTLY WORKING/PERSONAL PROJECT - OMNI-DIRECTIONAL ANKLE

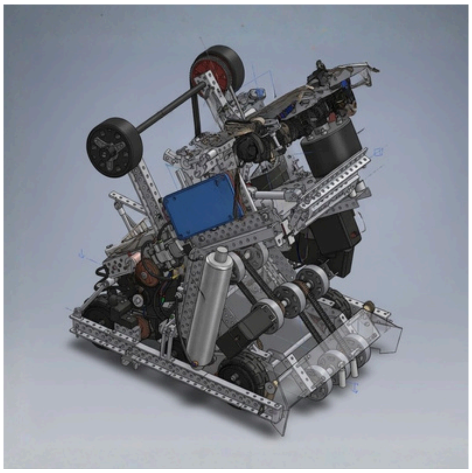




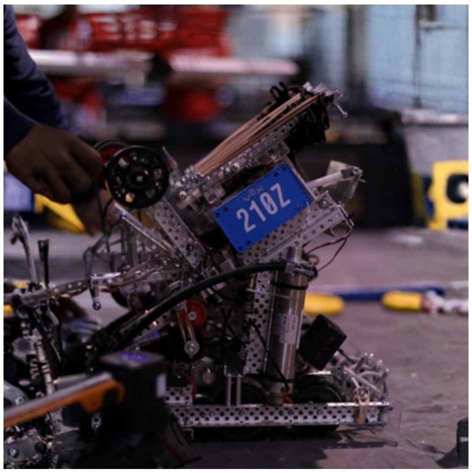
Still in progress of designing and will be created and built soon

- Currently designing and omni-directional prosthetic ankle, reflecting accurate angle and structure of a human ankle.
- Utilized a micro-controller within the ankle ball itself to simulate side to side move and angular movement

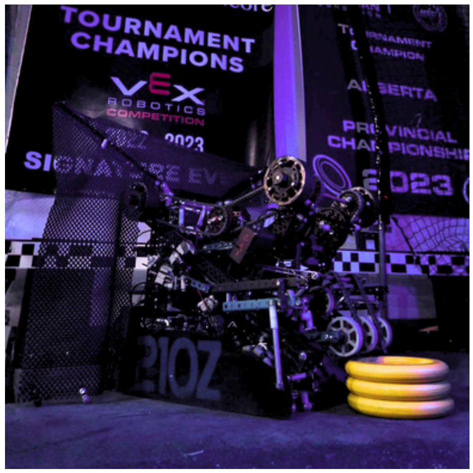
VRC SPIN UP- AUTONOMOUS FRISBEE GOLF ROBOT



- 3D modelled a mobile robot to intake, store and fire 3 foam discs into frisbee golf nets
- Designed a pneumatic piston loaded slingshot to expand and claim arena tiles post match
- Used a 600 RPM omnidirectional geared 4 motor drivetrain, balancing RPM vs torque trade offs

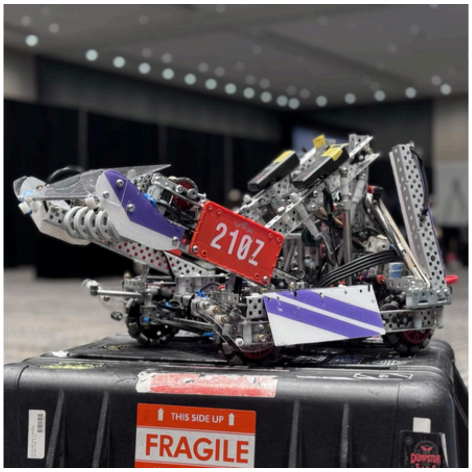


- Built 3600 RPM silicone flywheel to fire the discs in rapid succession. Used pneumatic pistons to create angle adjustable ramp.
- Used potentiometers, inertial sensors and tracking wheels to collect and score autonomously



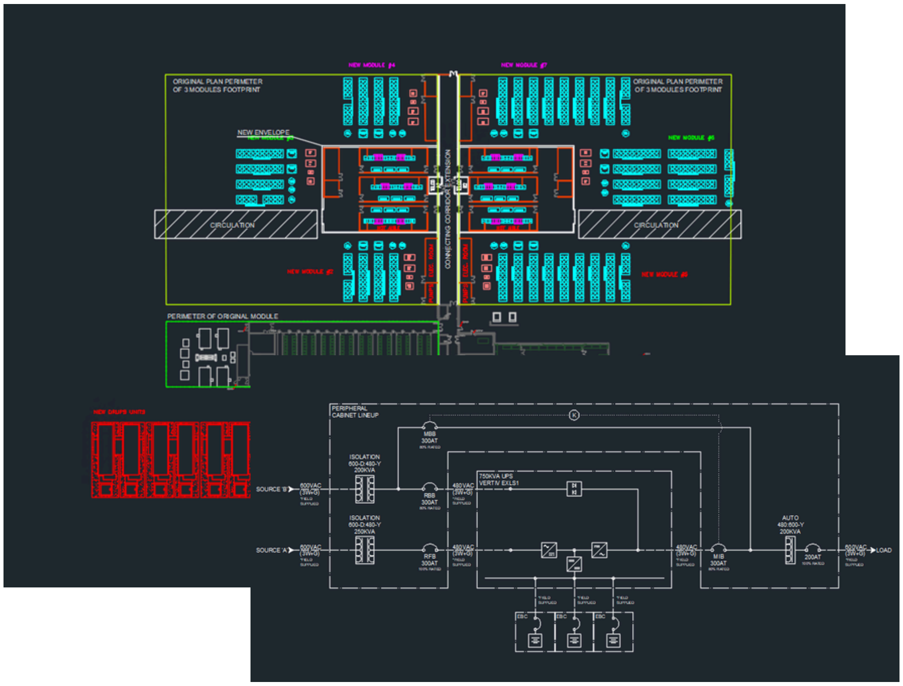
- Switched to piston powered catapult scoring after comparing design tests
- Modeled and tuned torque output vs. release angle to achieve consistent scoring arcs
- Applied trajectory physics to create a free-rotating disc base compared to the rotating arm (video in google album)

VRC OVER UNDER- TRIBALL SCORING AND CLIMBING ROBOT



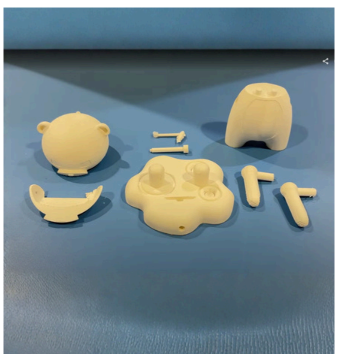
- Created a mobile tri-ball scoring and climbing robot with a 480 RPM drivetrain
- Built a catapult-based launcher before transitioning to a 360 RPM silicone flywheel system for improved shot consistency and cycle time
- Integrated pneumatic pistons to deploy a collapsible blocking mechanism and enable endgame climbing functionality to raise the robot from the field

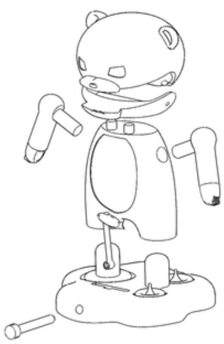
POWERHOUSE DATA CENTRE GROUP- AUTOCAD/THERMAL PROJECTS



- Used AutoCAD to develop electrical line diagrams of redundant data centre power systems, where incoming power is conditioned to remove surges, with excess energy rerouted to emergency backup batteries.
- Developed site diagrams for projects integrating thermal, power, and computing requirement rooms to optimize effeciency
- Completing thermal unit configuration and testing for 20kW to 500kW facilities, comparing dry-bulb temperatures with static pressure.

SYDE DESIGN COURSE- INTERACTIVE CANADIAN PUZZLE





- Designed an interactive assemblable Canadian themed kids toy for a 3D Design Course
- Utilized all Solidworks tools to create a multi-use toy, turn all arms, raising the fish and opening the jaw through a pin joint